

Plant Disease Detection Using CNN - Project Report (Updated)

This PowerPoint presentation outlines a student project on an AI system for identifying plant diseases using Convolutional Neural Networks (CNNs). It details the problem, methodology, implementation, and future enhancements for practical agricultural use.

Project Team

The project is developed by Team Crop Guardians: E23CSEU1206 - Kaushtubh Tiwari, E23CSEU1213 - Arnav Ahlawat.

Introduction

Agriculture is crucial to the global economy, but plant diseases reduce yields and farmer incomes. Manual identification is slow and error-prone, so the project proposes an AI system using CNNs to detect diseases from leaf images quickly and accurately.

Objectives

The system aims to develop an efficient ML-based tool for timely disease detection, improving crop yields, minimizing losses, and supporting sustainable farming practices.

Literature Review

Traditional methods rely on expert visual inspections, which are labor-intensive and inaccessible in remote areas. Early ML used SVM, KNN, and Random Forests with manual features, but they lacked scalability. Modern CNNs like AlexNet, VGGNet, and ResNet enable automatic feature extraction; this project addresses gaps in deployment, UI, and accessibility using PlantVillage dataset and full-stack tech.

Tech Stack

- Frontend: ReactJS
- Backend: NodeJS
- API: FastAPI (for ML model serving)

- Database: MongoDB
- ML: CNN with MobileNetV2 transfer learning on PlantVillage dataset.

Methodology

Data is sourced from PlantVillage, preprocessed, and split into train/test/validation sets. Transfer learning on MobileNetV2 (pre-trained on ImageNet) trains for 5 epochs. The model is tested, evaluated for accuracy, and saved for deployment.

System Flow

Users upload leaf images via drag-and-drop on the React homepage. FastAPI loads the model, predicts the disease, queries MongoDB for details, and displays results with a PDF download option.

Results

The presentation shows UI screenshots: localhost startup, prediction page, results page (with disease info), and PDF download. Model performance is evaluated via validation/test accuracy, though specific metrics are not detailed; images depict successful predictions.

Future Scope

Enhance accuracy with CNN features (Shallow VGG) plus Random Forest/XGBoost classifiers. Expand to more diseases by collecting additional datasets.